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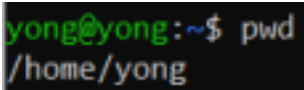
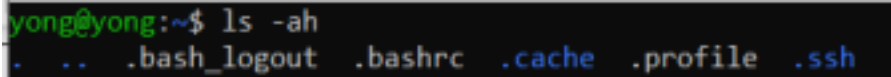
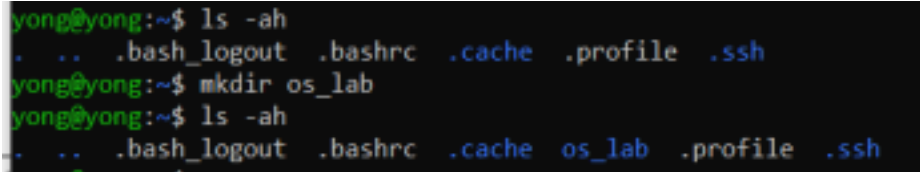
Student ID : A23CS0250

Section : 02

This lab assessment is designed to test your understanding and skills on some basic concepts and tools related to operating system. Please follow the instructions carefully and submit your answers in this word document and rename the file as **os-lab-assessment01-studentname-matricno.docx**.

ACTIVITY 1 : BASIC LINUX COMMANDS

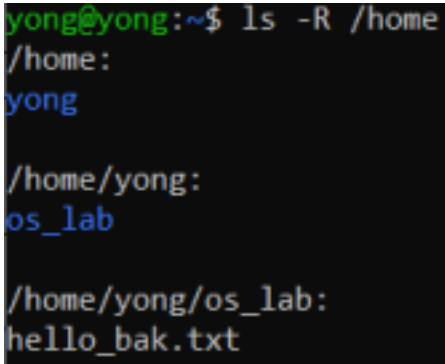
In this activity, you will practice some basic Linux commands that are useful for operating system tasks. For each question, write down the command you used and the output you got in your answer file. If the command does not produce any output, write "No output" instead.

1.	Display your current working directory.	
	Command	Output
	pwd	
2.	List all the files and directories in your home directory, including hidden ones.	
	Command	Output
	ls -ah	
3.	Create a new directory named "os_lab" in your home directory. List all the files and directories before and after running this command to show the changes.	
	Command	Output
	mkdir os_lab	

4.	Change your current working directory to "os_lab". Display your current location before and after running that command to show the changes.	
	Command	Output
	cd os_lab	<pre>yong@yong:~\$ pwd /home/yong yong@yong:~\$ cd os_lab yong@yong:~/os_lab\$ pwd /home/yong/os_lab</pre>
5.	Create a new file named "hello.txt" in "os_lab" and write "Hello, world!" in it. List all the files and directories before and after running this command to show the changes.	
	Command	Output
	nano hello.txt	<pre>yong@yong:~/os_lab\$ ls yong@yong:~/os_lab\$ nano hello.txt yong@yong:~/os_lab\$ ls hello.txt</pre>
6.	Display the content of "hello.txt".	
	Command	Output
	cat hello.txt	<pre>yong@yong:~/os_lab\$ cat hello.txt "Hello, world!"</pre>
7.	Copy "hello.txt" to another file named "hello_copy.txt" in the same directory. List all the files and directories before and after running this command to show the changes.	
	Command	Output

	cp hello.txt hello_copy.txt	<pre>yong@yong:~/os_lab\$ ls hello.txt yong@yong:~/os_lab\$ cp hello.txt hello_copy.txt yong@yong:~/os_lab\$ ls hello_copy.txt hello.txt</pre>
8.	Rename "hello_copy.txt" to "hello_bak.txt". List all the files and directories before and after running this command to show the changes.	
	Command	Output
	mv hello_copy.txt hello_bak.txt	<pre>yong@yong:~/os_lab\$ ls hello_copy.txt hello.txt yong@yong:~/os_lab\$ mv hello_copy.txt hello_bak.txt yong@yong:~/os_lab\$ ls hello_bak.txt hello.txt</pre>

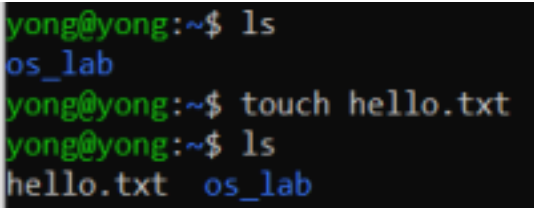
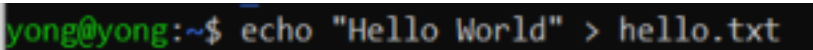
9.	Delete "hello.txt". List all the files and directories before and after running this command to show the changes.	
	Command	Output
	rm hello.txt	<pre>yong@yong:~/os_lab\$ ls hello_bak.txt hello.txt yong@yong:~/os_lab\$ rm hello.txt yong@yong:~/os_lab\$ ls hello_bak.txt</pre>
10.	Display your home directory with additional information, together with subdirectories contents.	
	Command	Output

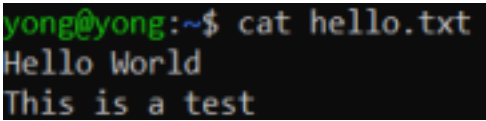
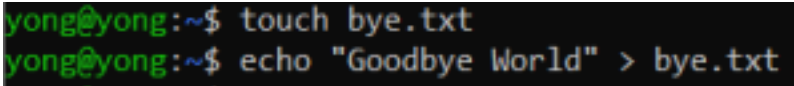
	ls -R /home	
	Total Mark:	

ACTIVITY 2 : Text File Manipulation

In this activity, you will practice some commands to create and manipulate text files using echo, cat and touch.

For each question, write down the command and the output (if any) in the answer sheet.

1.	Create an empty file named "hello.txt" in the current working directory. List all the files and directories before and after running this command to show the changes.	
	Command	Output
	touch hello.txt	
2.	Write the text "Hello World" to "hello.txt" using echo.	
	Command	Output
	echo "Hello World" > hello.txt	

3.	Append the text "This is a test" to "hello.txt" using echo.	
	Command	Output
	echo "This is a test" >> hello.txt	
4.	Display the contents of "hello.txt" using cat.	
	Command	Output
	cat hello.txt	
5.	Create another file named "bye.txt" with the text "Goodbye World" using echo.	
	Command	Output
	touch bye.txt echo "Goodbye World" > bye.txt	

6.	Concatenate the contents of "hello.txt" and "bye.txt" and display them using cat.	
	Command	Output
	cat hello.txt bye.txt	
7.	Concatenate the contents of "hello.txt" and "bye.txt" and write them to a new file named "greeting.txt" using cat.	

	Command	Output
	cat hello.txt bye.txt > greeting.txt	
8.	Display the contents of "greeting.txt" using cat.	
	Command	Output
	cat greeting.txt	
9.	Edit the file "greeting.txt" using Nano and change the word "World" to your name in both lines.	
	Command	Output
	nano greeting.txt	
10.	Display the modified contents of "greeting.txt" using cat.	
	Command	Output
	cat greeting.txt	
	Total Mark:	

ACTIVITY 3 : WRITE, COMPILE AND RUN C PROGRAM

In this activity, you will practice how to write, compile and run a simple C program using gcc. For each question, write down the command and the output (if any) in the answer sheet.

1.	Write a C program that prints "Hello World" to the standard output using Nano and save it as "hello.c".	
	Command	Output
	nano hello.c	
2.	Compile the program "hello.c" using gcc and generate an executable file named "hello". List all the files and directories before and after running this command to show the changes.	
	Command	Output
	gcc hello.c - o hello	
3.	Run the executable file "hello" and display its output.	
	Command	Output
	./hello	

4.	Write a C program that takes two integers as command line arguments and prints their sum to the standard output using Nano and save it as "sum.c".	
	Command	Output
	nano sum.c	
5.	Compile the program "sum.c" using gcc and generate an executable file named "sum". List all the files and directories before and after running this command to show the changes.	
	Command	Output
	gcc sum.c - o sum	
6.	Run the executable file "sum" with 10 and 20 as arguments and display its output.	
	Command	Output
	./sum	

7.	Run the executable file "sum" with -5 and 15 as arguments and display its output.	
	Command	Output
	./sum	

8.	Write a C program that reads a line of text from the standard input and prints it to the standard output using Nano and save it as "echo.c".	
	Command	Output
	nano echo.c	
9.	Compile the program "echo.c" using gcc and generate an executable file named "echo". List all the files and directories before and after running this command to show the changes.	
	Command	Output
	gcc echo.c - o echo	
10.	Run the executable file "echo", type "Hello World" as input and display its output.	

	Command	Output
	./echo	
	Total Mark:	